

Electronics of Noise Control

When an electronically produced inverse wave is added to the original unwanted sound, the result is sound cancellation. This method of noise control is becoming increasingly popular for a variety of uses



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The word 'noise' comes from the Latin word 'nausea' meaning 'seasickness.' Noise pollution is a displeasing human-, animal- or machine-created sound that disrupts the activity or balance of human or animal life. It is a type of energy pollution in which distracting, irritating or damaging sounds are freely audible. As with other forms of energy pollution (such as heat and light pollution), noise pollution contaminants are not physical particles, but rather waves

that interfere with naturally-occurring waves of a similar type in the same environment. Thus the definition of noise pollution is open to debate, and there is no clear border as to which sounds may constitute noise pollution.

In the narrowest sense, sounds are considered noise pollution if they adversely affect wildlife, human activity or are capable of damaging physical structures on a regular, repeating basis. In the broadest sense of the term, a sound may be considered noise pollution if it disturbs any natural process or harms humans even if the sound does

not occur on a regular basis.

Sources of noise pollution

There are listless indoor and outdoor noise-pollution sources. Noise pollution, like other pollutants, is a byproduct of industrialisation, urbanisation and modern civilisation. Broadly speaking, it has two sources: industrial and non-industrial. Industrial sources include the noise from various industries and big machines working at a very high speed and high noise intensity. Non-industrial sources of noise include the noise created by transport/vehicular traffic and the neighbourhood activities.

Noise-pollution sources can also be divided into natural and manmade categories. Major noise sources are road traffic, aircrafts, railroads, construction, industries, buildings and consumer products.

Health effects

Noise, the unwanted sound, can damage physiological and psychological health. Noise pollution can cause annoyance and aggression, hypertension, high stress levels, tinnitus, hearing loss, sleep disturbances and other harmful effects. Stress and hypertension are the leading causes of health problems, whereas tinnitus can lead to forgetfulness, severe depression and at times panic attacks.

Chronic exposure to noise may cause noise-induced hearing loss. High noise levels can contribute to cardiovascular effects and exposure to moderately high levels during a single eight-hour period causes a statistical rise of five to ten points in blood pres-